

Code Converters

Large variety of codes are available, which results in the use of different codes by different digital systems.

Sometimes, o/p of one system may be used as the i/p to another. $\therefore$, a conversion circuit must be inserted b/w the two systems.

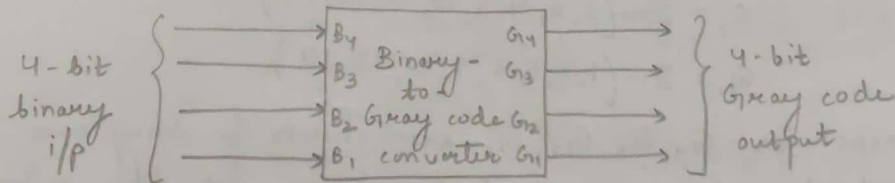
A code converter is a logic circuit that changes data presented in one type of binary code to another type of binary code.

The following are some of most commonly used code converters:-

- (i) BCD - to - binary
- (ii) Binary - to - BCD
- (iii) Binary - to - Gray code
- (iv) Gray code - to - binary
- (v) ASCII - to - EBCDIC
- (vi) EBCDIC - to - ASCII
- (vii) BCD - to - XS-3

Design of a 4-bit Binary-to-Gray Code Converter -

The i/p to the 4-bit binary-to-Gray code converter is a 4-bit binary & o/p is a 4-bit Gray code. The block diagram is shown below:-



Block Diagram

There are 16 possible combinations of 4-bit binary i/p & all of them are valid. Hence no don't cares.

The 4-bit binary & the corresponding Gray code are shown in the conversion table (or truth table) shown below:-

4-bit binary				4-bit Gray			
B_4	B_3	B_2	B_1	G_4	G_3	G_2	G_1
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	0	0	0	1	1
0	0	1	1	0	0	1	0
0	1	0	0	0	1	1	0
0	1	0	1	0	1	1	1
0	1	1	0	0	1	0	1
0	1	1	1	0	1	0	0
1	0	0	0	1	1	0	0
1	0	0	1	1	1	0	1
1	0	1	0	1	1	1	1
1	0	1	1	1	1	1	0
1	1	0	0	1	0	1	0
1	1	0	1	1	0	1	1
1	1	1	0	1	0	0	1
1	1	1	1	1	0	0	0

The expressions for the Gray code o/p can be written as:-

$$G_4 = \sum m(8, 9, 10, 11, 12, 13, 14, 15)$$

$$G_3 = \sum m(4, 5, 6, 7, 8, 9, 10, 11)$$

$$G_2 = \sum m(2, 3, 4, 5, 10, 11, 12, 13)$$

$$G_1 = \sum m(1, 2, 5, 6, 9, 10, 13, 14)$$

The k-maps for G_4, G_3, G_2, G_1 are shown b they can be simplified to obtain minimal expressions for the outputs:-

$B_4 B_3$	$B_2 B_1$			
	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

$$G_4 = B_4$$

$B_4 B_3$	$B_2 B_1$			
	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

$$G_3 = \bar{B}_4 B_3 + B_4 \bar{B}_3$$

$$G_3 = B_4 \oplus B_3$$

$B_4 B_3$	$B_2 B_1$	00	01	11	10
00		0	1	3	2
01		4	5	7	6
11		12	13	15	14
10		8	9	11	10

$$G_2 = B_3 \bar{B}_2 + \bar{B}_3 B_2$$

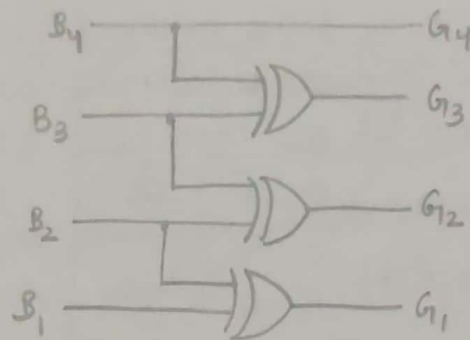
$$G_2 = B_3 \oplus B_2$$

$B_4 B_3$	$B_2 B_1$	00	01	11	10
00		0	1	3	2
01		4	5	7	6
11		12	13	15	14
10		8	9	11	10

$$G_1 = \bar{B}_2 B_1 + B_2 \bar{B}_1$$

$$G_1 = B_2 \oplus B_1$$

The above expressions can be implemented using ex-OR gates as shown :-

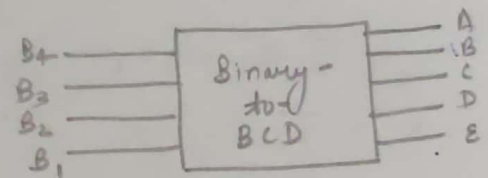


Logic diagram

Similarly, implement Gray-to-Binary code converter.

Design of 4-bit Binary-to-BCD code converter -

Decimal	4-bit binary				BCD o/p				
	B_4	B_3	B_2	B_1	A	B	C	D	E
0	0	0	0	0	0	0	0	0	0
1	0	0	0	1	0	0	0	0	1
2	0	0	1	0	0	0	0	1	0
3	0	0	1	1	0	0	0	1	1
4	0	1	0	0	0	0	1	0	0
5	0	1	0	1	0	0	1	0	1
6	0	1	1	0	0	0	1	1	0
7	0	1	1	1	0	0	1	1	1
8	1	0	0	0	0	1	0	0	0
9	1	0	0	1	0	1	0	0	1
10	1	0	1	0	1	0	0	0	0
11	1	0	1	1	1	0	0	0	1
12	1	1	0	0	1	0	0	1	0
13	1	1	0	1	1	0	0	1	1
14	1	1	1	0	1	0	1	0	0
15	1	1	1	1	1	0	1	0	1



$$A = \sum m(10, 11, 12, 13, 14, 15)$$

$$B = \sum m(8, 9)$$

$$C = \sum m(4, 5, 6, 7, 14, 15)$$

$$D = \sum m(2, 3, 6, 7, 12, 13)$$

$$E = \sum m(1, 3, 5, 7, 9, 11, 13, 15)$$

Using k-map simplification,

$$A = B_4 B_3 + B_4 B_2$$

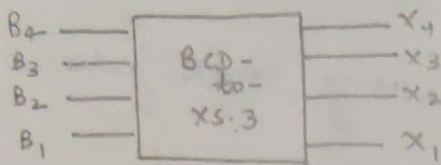
$$B = B_4 \overline{B_3} \overline{B_2}$$

$$C = \overline{B_4} B_3 + B_3 B_2$$

$$D = B_4 B_3 \overline{B_2} + \overline{B_4} B_2$$

$$E = B_1$$

Design of a 4-bit BCD-to-XS-3 Code Converter -



B_4	B_3	B_2	B_1	X_4	X_3	X_2	X_1
0	0	0	0	0	0	1	1
0	0	0	1	0	1	0	0
0	0	1	0	0	1	0	1
0	0	1	1	0	1	1	0
0	1	0	0	0	1	1	1
0	1	0	1	1	0	0	0
0	1	1	0	1	0	0	1
0	1	1	1	1	0	1	0
1	0	0	0	1	0	1	1
1	0	0	1	1	1	0	0

$$X_4 = \sum m(5, 6, 7, 8, 9) + d(10, 11, 12, 13, 14, 15)$$

$$X_3 = \sum m(1, 2, 3, 4, 9) + d(10, 11, 12, 13, 14, 15)$$

$$X_2 = \sum m(0, 3, 4, 7, 8) + d(10, 11, 12, 13, 14, 15)$$

$$X_1 = \sum m(0, 2, 4, 6, 8) + d(10, 11, 12, 13, 14, 15)$$

Solving using k-map,

$$X_4 = B_4 + B_3 B_2 + B_3 B_1$$

$$X_3 = B_3 \overline{B_2} \overline{B_1} + \overline{B_3} B_1 + \overline{B_3} B_2$$

$$X_2 = \overline{B_2} \overline{B_1} + B_2 B_1$$

$$X_1 = \overline{B_1}$$

A Magnitude
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